

LIFEWS AgriShine(TM) GrowMeal(TM)

Primary 4-6 ASSESSMENT & QUIZ SYSTEM

TEACHER MARKING & MODERATION GUIDE

Purpose: assess knowledge, practical competence, investigation/data skills, and mission-based problem solving without reducing garden learning to recall alone.

Domain	Weight	Primary evidence
Knowledge	20%	Oral/written response, concept explanation
Practical competence	25%	Demonstration, technique, safety, record quality
Investigation/Data	30%	Measurement, table/graph, calculation, interpretation
Mission problem solving	25%	Diagnosis, design, recommendation, evidence-based decision

Performance rubric for open/practical items

Score	Descriptor
3 - Proficient	Method is appropriate, evidence is accurate/traceable, and explanation or decision is supported.
2 - Developing	Method is mostly appropriate with minor errors or incomplete reasoning.
1 - Beginning	Major support is needed; evidence or reasoning is incomplete.
0 - Not demonstrated	No valid evidence or the task is not completed.

Moderation rule: award credit for scientifically or practically valid alternatives when they satisfy the task. Do not award marks for unsupported claims. For Nursery, use oral responses and observed performance where writing is not developmentally appropriate.

ANSWER & MARKING GUIDE 01 - Raised Food Garden

AQ-201 - Knowledge

Define or explain one core idea related to raised-bed crop growth and spacing.

Expected answer / marking basis: Accurate level-appropriate explanation.

AQ-202 - Knowledge

Why is a fair comparison important when studying the raised bed?

Expected answer / marking basis: It reduces bias and supports a valid comparison.

AQ-203 - Knowledge

What should be kept the same when comparing two conditions?

Expected answer / marking basis: Relevant controlled variables.

AQ-204 - Practical

Use an appropriate tool to collect a repeatable measurement.

Expected answer / marking basis: 0-3 technique/units rubric.

AQ-205 - Practical

Show one quality or safety check before recording the result.

Expected answer / marking basis: 0-3 QC/safety rubric.

AQ-206 - Investigation/Data

Collect at least three values and calculate a useful summary such as mean, range, rate or percentage.

Expected answer / marking basis: Correct calculation with units.

AQ-207 - Investigation/Data

Graph or tabulate the result and identify the main pattern.

Expected answer / marking basis: Accurate graph/table + defensible pattern.

AQ-208 - Mission

Identify one limitation in your investigation and explain how it could affect the result.

Expected answer / marking basis: Real limitation + plausible effect.

AQ-209 - Mission

Recommend one management or design change based on the evidence.

Expected answer / marking basis: Evidence-linked recommendation.

AQ-210 - Mission

Design a follow-up test that would strengthen the conclusion.

Expected answer / marking basis: Testable, safe follow-up with clear variable/metric.

ANSWER & MARKING GUIDE 02 - Geometry Garden

AQ-211 - Knowledge

Define or explain one core idea related to shapes, sides and garden spaces.

Expected answer / marking basis: Accurate level-appropriate explanation.

AQ-212 - Knowledge

Why is a fair comparison important when studying the garden bed or path?

Expected answer / marking basis: It reduces bias and supports a valid comparison.

AQ-213 - Knowledge

What should be kept the same when comparing two conditions?

Expected answer / marking basis: Relevant controlled variables.

AQ-214 - Practical

Use an appropriate tool to collect a repeatable shape or length.

Expected answer / marking basis: 0-3 technique/units rubric.

AQ-215 - Practical

Show one quality or safety check before recording the result.

Expected answer / marking basis: 0-3 QC/safety rubric.

AQ-216 - Investigation/Data

Collect at least three values and calculate a useful summary such as mean, range, rate or percentage.

Expected answer / marking basis: Correct calculation with units.

AQ-217 - Investigation/Data

Graph or tabulate the result and identify the main pattern.

Expected answer / marking basis: Accurate graph/table + defensible pattern.

AQ-218 - Mission

Identify one limitation in your investigation and explain how it could affect the result.

Expected answer / marking basis: Real limitation + plausible effect.

AQ-219 - Mission

Recommend one management or design change based on the evidence.

Expected answer / marking basis: Evidence-linked recommendation.

AQ-220 - Mission

Design a follow-up test that would strengthen the conclusion.

Expected answer / marking basis: Testable, safe follow-up with clear variable/metric.

ANSWER & MARKING GUIDE 03 - Grid Experiment Garden

AQ-221 - Knowledge

Define or explain one core idea related to treatment-control comparisons and fair testing.

Expected answer / marking basis: Accurate level-appropriate explanation.

AQ-222 - Knowledge

Why is a fair comparison important when studying the grid cell?

Expected answer / marking basis: It reduces bias and supports a valid comparison.

AQ-223 - Knowledge

What should be kept the same when comparing two conditions?

Expected answer / marking basis: Relevant controlled variables.

AQ-224 - Practical

Use an appropriate tool to collect a repeatable treatment response.

Expected answer / marking basis: 0-3 technique/units rubric.

AQ-225 - Practical

Show one quality or safety check before recording the result.

Expected answer / marking basis: 0-3 QC/safety rubric.

AQ-226 - Investigation/Data

Collect at least three values and calculate a useful summary such as mean, range, rate or percentage.

Expected answer / marking basis: Correct calculation with units.

AQ-227 - Investigation/Data

Graph or tabulate the result and identify the main pattern.

Expected answer / marking basis: Accurate graph/table + defensible pattern.

AQ-228 - Mission

Identify one limitation in your investigation and explain how it could affect the result.

Expected answer / marking basis: Real limitation + plausible effect.

AQ-229 - Mission

Recommend one management or design change based on the evidence.

Expected answer / marking basis: Evidence-linked recommendation.

AQ-230 - Mission

Design a follow-up test that would strengthen the conclusion.

Expected answer / marking basis: Testable, safe follow-up with clear variable/metric.

ANSWER & MARKING GUIDE 04 - Keyhole Garden

AQ-231 - Knowledge

Define or explain one core idea related to access, moisture and resource flow in keyhole beds.

Expected answer / marking basis: Accurate level-appropriate explanation.

AQ-232 - Knowledge

Why is a fair comparison important when studying the keyhole bed?

Expected answer / marking basis: It reduces bias and supports a valid comparison.

AQ-233 - Knowledge

What should be kept the same when comparing two conditions?

Expected answer / marking basis: Relevant controlled variables.

AQ-234 - Practical

Use an appropriate tool to collect a repeatable system observation.

Expected answer / marking basis: 0-3 technique/units rubric.

AQ-235 - Practical

Show one quality or safety check before recording the result.

Expected answer / marking basis: 0-3 QC/safety rubric.

AQ-236 - Investigation/Data

Collect at least three values and calculate a useful summary such as mean, range, rate or percentage.

Expected answer / marking basis: Correct calculation with units.

AQ-237 - Investigation/Data

Graph or tabulate the result and identify the main pattern.

Expected answer / marking basis: Accurate graph/table + defensible pattern.

AQ-238 - Mission

Identify one limitation in your investigation and explain how it could affect the result.

Expected answer / marking basis: Real limitation + plausible effect.

AQ-239 - Mission

Recommend one management or design change based on the evidence.

Expected answer / marking basis: Evidence-linked recommendation.

AQ-240 - Mission

Design a follow-up test that would strengthen the conclusion.

Expected answer / marking basis: Testable, safe follow-up with clear variable/metric.

ANSWER & MARKING GUIDE 05 - Compost Garden

AQ-241 - Knowledge

Define or explain one core idea related to temperature, moisture, volume and maturity.

Expected answer / marking basis: Accurate level-appropriate explanation.

AQ-242 - Knowledge

Why is a fair comparison important when studying the compost pile?

Expected answer / marking basis: It reduces bias and supports a valid comparison.

AQ-243 - Knowledge

What should be kept the same when comparing two conditions?

Expected answer / marking basis: Relevant controlled variables.

AQ-244 - Practical

Use an appropriate tool to collect a repeatable compost indicator.

Expected answer / marking basis: 0-3 technique/units rubric.

AQ-245 - Practical

Show one quality or safety check before recording the result.

Expected answer / marking basis: 0-3 QC/safety rubric.

AQ-246 - Investigation/Data

Collect at least three values and calculate a useful summary such as mean, range, rate or percentage.

Expected answer / marking basis: Correct calculation with units.

AQ-247 - Investigation/Data

Graph or tabulate the result and identify the main pattern.

Expected answer / marking basis: Accurate graph/table + defensible pattern.

AQ-248 - Mission

Identify one limitation in your investigation and explain how it could affect the result.

Expected answer / marking basis: Real limitation + plausible effect.

AQ-249 - Mission

Recommend one management or design change based on the evidence.

Expected answer / marking basis: Evidence-linked recommendation.

AQ-250 - Mission

Design a follow-up test that would strengthen the conclusion.

Expected answer / marking basis: Testable, safe follow-up with clear variable/metric.

ANSWER & MARKING GUIDE 06 - Gravity Drip Garden

AQ-251 - Knowledge

Define or explain one core idea related to flow rate, uniformity and applied water.

Expected answer / marking basis: Accurate level-appropriate explanation.

AQ-252 - Knowledge

Why is a fair comparison important when studying the drip emitter?

Expected answer / marking basis: It reduces bias and supports a valid comparison.

AQ-253 - Knowledge

What should be kept the same when comparing two conditions?

Expected answer / marking basis: Relevant controlled variables.

AQ-254 - Practical

Use an appropriate tool to collect a repeatable flow rate.

Expected answer / marking basis: 0-3 technique/units rubric.

AQ-255 - Practical

Show one quality or safety check before recording the result.

Expected answer / marking basis: 0-3 QC/safety rubric.

AQ-256 - Investigation/Data

Collect at least three values and calculate a useful summary such as mean, range, rate or percentage.

Expected answer / marking basis: Correct calculation with units.

AQ-257 - Investigation/Data

Graph or tabulate the result and identify the main pattern.

Expected answer / marking basis: Accurate graph/table + defensible pattern.

AQ-258 - Mission

Identify one limitation in your investigation and explain how it could affect the result.

Expected answer / marking basis: Real limitation + plausible effect.

AQ-259 - Mission

Recommend one management or design change based on the evidence.

Expected answer / marking basis: Evidence-linked recommendation.

AQ-260 - Mission

Design a follow-up test that would strengthen the conclusion.

Expected answer / marking basis: Testable, safe follow-up with clear variable/metric.

ANSWER & MARKING GUIDE 07 - Vertical Garden

AQ-261 - Knowledge

Define or explain one core idea related to performance across vertical levels.

Expected answer / marking basis: Accurate level-appropriate explanation.

AQ-262 - Knowledge

Why is a fair comparison important when studying the vertical planter?

Expected answer / marking basis: It reduces bias and supports a valid comparison.

AQ-263 - Knowledge

What should be kept the same when comparing two conditions?

Expected answer / marking basis: Relevant controlled variables.

AQ-264 - Practical

Use an appropriate tool to collect a repeatable level response.

Expected answer / marking basis: 0-3 technique/units rubric.

AQ-265 - Practical

Show one quality or safety check before recording the result.

Expected answer / marking basis: 0-3 QC/safety rubric.

AQ-266 - Investigation/Data

Collect at least three values and calculate a useful summary such as mean, range, rate or percentage.

Expected answer / marking basis: Correct calculation with units.

AQ-267 - Investigation/Data

Graph or tabulate the result and identify the main pattern.

Expected answer / marking basis: Accurate graph/table + defensible pattern.

AQ-268 - Mission

Identify one limitation in your investigation and explain how it could affect the result.

Expected answer / marking basis: Real limitation + plausible effect.

AQ-269 - Mission

Recommend one management or design change based on the evidence.

Expected answer / marking basis: Evidence-linked recommendation.

AQ-270 - Mission

Design a follow-up test that would strengthen the conclusion.

Expected answer / marking basis: Testable, safe follow-up with clear variable/metric.

ANSWER & MARKING GUIDE 08 - Crop Rotation Garden

AQ-271 - Knowledge

Define or explain one core idea related to crop-family sequence and soil cover.

Expected answer / marking basis: Accurate level-appropriate explanation.

AQ-272 - Knowledge

Why is a fair comparison important when studying the rotation record?

Expected answer / marking basis: It reduces bias and supports a valid comparison.

AQ-273 - Knowledge

What should be kept the same when comparing two conditions?

Expected answer / marking basis: Relevant controlled variables.

AQ-274 - Practical

Use an appropriate tool to collect a repeatable rotation interval.

Expected answer / marking basis: 0-3 technique/units rubric.

AQ-275 - Practical

Show one quality or safety check before recording the result.

Expected answer / marking basis: 0-3 QC/safety rubric.

AQ-276 - Investigation/Data

Collect at least three values and calculate a useful summary such as mean, range, rate or percentage.

Expected answer / marking basis: Correct calculation with units.

AQ-277 - Investigation/Data

Graph or tabulate the result and identify the main pattern.

Expected answer / marking basis: Accurate graph/table + defensible pattern.

AQ-278 - Mission

Identify one limitation in your investigation and explain how it could affect the result.

Expected answer / marking basis: Real limitation + plausible effect.

AQ-279 - Mission

Recommend one management or design change based on the evidence.

Expected answer / marking basis: Evidence-linked recommendation.

AQ-280 - Mission

Design a follow-up test that would strengthen the conclusion.

Expected answer / marking basis: Testable, safe follow-up with clear variable/metric.

ANSWER & MARKING GUIDE 09 - Water-Smart Garden

AQ-281 - Knowledge

Define or explain one core idea related to infiltration, irrigation and water productivity.

Expected answer / marking basis: Accurate level-appropriate explanation.

AQ-282 - Knowledge

Why is a fair comparison important when studying the water-smart bed?

Expected answer / marking basis: It reduces bias and supports a valid comparison.

AQ-283 - Knowledge

What should be kept the same when comparing two conditions?

Expected answer / marking basis: Relevant controlled variables.

AQ-284 - Practical

Use an appropriate tool to collect a repeatable water metric.

Expected answer / marking basis: 0-3 technique/units rubric.

AQ-285 - Practical

Show one quality or safety check before recording the result.

Expected answer / marking basis: 0-3 QC/safety rubric.

AQ-286 - Investigation/Data

Collect at least three values and calculate a useful summary such as mean, range, rate or percentage.

Expected answer / marking basis: Correct calculation with units.

AQ-287 - Investigation/Data

Graph or tabulate the result and identify the main pattern.

Expected answer / marking basis: Accurate graph/table + defensible pattern.

AQ-288 - Mission

Identify one limitation in your investigation and explain how it could affect the result.

Expected answer / marking basis: Real limitation + plausible effect.

AQ-289 - Mission

Recommend one management or design change based on the evidence.

Expected answer / marking basis: Evidence-linked recommendation.

AQ-290 - Mission

Design a follow-up test that would strengthen the conclusion.

Expected answer / marking basis: Testable, safe follow-up with clear variable/metric.

ANSWER & MARKING GUIDE 10 - Intro Soilless Garden

AQ-291 - Knowledge

Define or explain one core idea related to flow, pH, EC and root-zone condition.

Expected answer / marking basis: Accurate level-appropriate explanation.

AQ-292 - Knowledge

Why is a fair comparison important when studying the soilless system?

Expected answer / marking basis: It reduces bias and supports a valid comparison.

AQ-293 - Knowledge

What should be kept the same when comparing two conditions?

Expected answer / marking basis: Relevant controlled variables.

AQ-294 - Practical

Use an appropriate tool to collect a repeatable system reading.

Expected answer / marking basis: 0-3 technique/units rubric.

AQ-295 - Practical

Show one quality or safety check before recording the result.

Expected answer / marking basis: 0-3 QC/safety rubric.

AQ-296 - Investigation/Data

Collect at least three values and calculate a useful summary such as mean, range, rate or percentage.

Expected answer / marking basis: Correct calculation with units.

AQ-297 - Investigation/Data

Graph or tabulate the result and identify the main pattern.

Expected answer / marking basis: Accurate graph/table + defensible pattern.

AQ-298 - Mission

Identify one limitation in your investigation and explain how it could affect the result.

Expected answer / marking basis: Real limitation + plausible effect.

AQ-299 - Mission

Recommend one management or design change based on the evidence.

Expected answer / marking basis: Evidence-linked recommendation.

AQ-300 - Mission

Design a follow-up test that would strengthen the conclusion.

Expected answer / marking basis: Testable, safe follow-up with clear variable/metric.

LEVEL MODERATION & REPORTING

Recommended reporting: maintain the four domain scores separately as well as a weighted total. A student may have a strong overall total but still need targeted support in practical technique, data interpretation or problem solving.

Suggested achievement bands: 85-100 Advanced; 70-84 Secure; 55-69 Developing; 40-54 Emerging; below 40 Requires targeted support. Schools may adjust bands to their approved assessment policy.

Teacher moderation notes: